Oluwatosin Ayeni

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12 November 2026

Inworld AI

Dear Hiring Team,

I am writing to apply for the Staff / Principal Machine Learning Engineer (Serving/Inference) role at Inworld AI. The clearest signal that this role is right for me: I have already built and benchmarked the exact technology stack it demands — vLLM, SGLang, TensorRT-LLM, and CUDA-optimised inference pipelines on H100 and RTX 4090 hardware — and I am ready to take those learnings into Inworld's production realtime serving environment at scale.

Three achievements that map directly to this role:

- **H100 benchmarking at production scale:** On dual NVIDIA H100 NVL 94GB (NVLink 4.0, CUDA 13.4) I benchmarked vLLM v0.30 vs SGLang v0.5.20 serving Qwen3-8B BF16 with TP=2. SGLang's RadixAttention prefix caching delivered a **1,443× TTFT speedup** at 131K-token context. I built the live results dashboard at inference-web-theta.vercel.app/benchmarks/llm — this is the level of rigour I bring to serving infrastructure decisions.
- **Multimodal inference on vLLM-Omni:** My *lab-vllm-omni* project built a phased inference lab across all five output modalities — text, audio, image, video, and structured output. Inworld serves speech models and LLMs together in a realtime consumer-facing loop; I have worked through the exact latency and scheduling tradeoffs that makes hard.
- **Production serving from cloud to Kubernetes:** My *llm-inference-platform* deploys vLLM + SGLang + LiteLLM gateway on RunPod and Vast.ai with KEDA autoscaling, Prometheus/Grafana/Loki/AlertManager observability, and Velero backup — a complete production inference stack, not a demo.

Inworld AI sits at the intersection of realtime AI and consumer-grade low latency — building first-party speech models and serving multimodal LLMs as modular APIs for high-volume workloads. That combination of speech, LLM, and production reliability is exactly the class of problem I have been building toward. The latency constraints on a voice-first AI product are unforgiving, and I am drawn to working where the gap between a good model and a good user experience is almost entirely a serving engineering problem.

I enjoy working in small, technically deep teams where shipping reliable systems matters more than process. I look forward to the possibility of discussing how I can contribute to Inworld's inference infrastructure.

Yours sincerely,

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